

Experiment 26: Fruit and Its Colour Using Backtracking in Prolog

Aim

To implement a **Prolog program to represent fruits and their colours** and demonstrate the concept of **backtracking** to obtain multiple solutions.

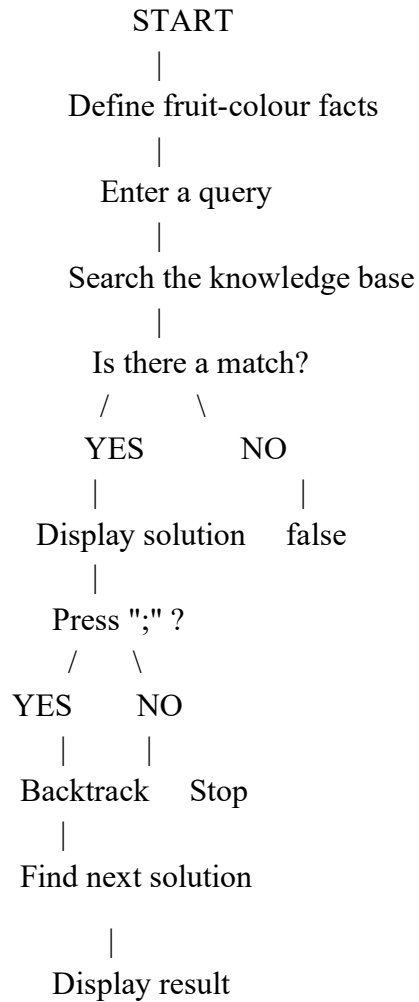
Objective

1. To represent fruits and their colours using Prolog facts.
2. To understand the concept of backtracking in Prolog.
3. To retrieve the colour of a specific fruit.
4. To find all fruits having a particular colour.
5. To obtain multiple solutions using the ; operator.

Algorithm

1. Start the program.
2. Define facts containing the names of fruits and their colours.
3. Enter a query with a variable to search for a fruit or colour.
4. Prolog searches the database from top to bottom.
5. Prolog returns the first matching solution.
6. Press ; to request another solution.
7. Prolog backtracks to the previous choice point and searches for the next matching fact.
8. Continue until all possible solutions are found.
9. Display the results.
10. Stop.

Flowchart



Prolog Code

% Fruit and Colour Database

```
fruit(apple, red).
fruit(banana, yellow).
fruit(orange, orange).
fruit(grapes, green).
fruit(mango, yellow).
fruit(strawberry, red).
fruit(guava, green).
```

Sample Output

Query 1: Find the colour of Apple

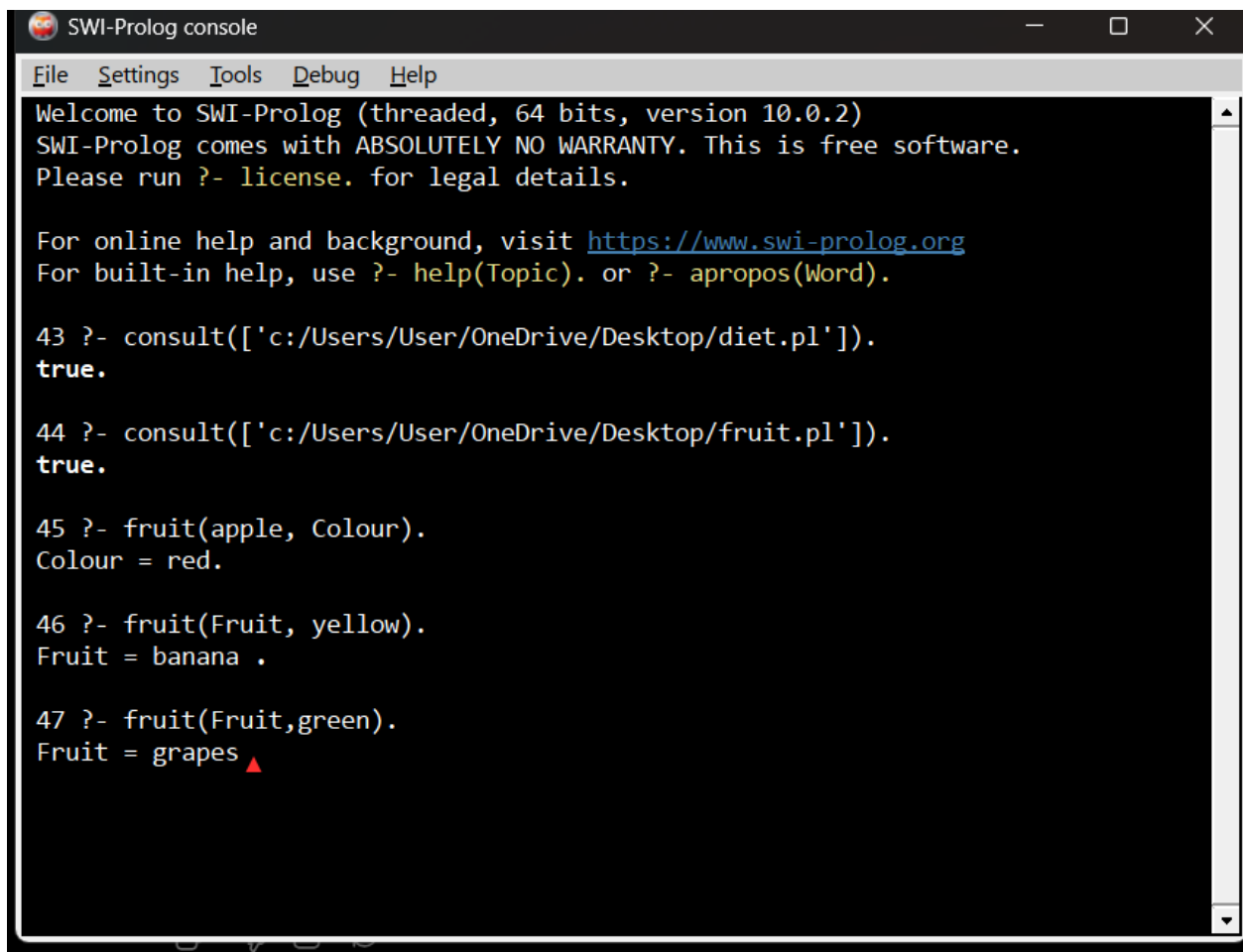
```
?- fruit(apple, Colour).
```

Query 2: Find all fruits and their colours using backtracking

```
?- fruit(Fruit, Colour).
```

Query 3: Find all yellow fruits

```
?- fruit(Fruit, yellow).
```

A screenshot of the SWI-Prolog console window. The window has a title bar with the text "SWI-Prolog console" and standard window controls (minimize, maximize, close). Below the title bar is a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The main area of the window is a black terminal with white text. It displays the following content:
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run `?- license.` for legal details.

For online help and background, visit <https://www.swi-prolog.org>
For built-in help, use `?- help(Topic).` or `?- apropos(Word).`

43 `?- consult(['c:/Users/User/OneDrive/Desktop/diet.pl']).`
`true.`

44 `?- consult(['c:/Users/User/OneDrive/Desktop/fruit.pl']).`
`true.`

45 `?- fruit(apple, Colour).`
`Colour = red.`

46 `?- fruit(Fruit, yellow).`
`Fruit = banana .`

47 `?- fruit(Fruit,green).`
`Fruit = grapes` followed by a red triangle cursor.
At the bottom of the window, there are three small icons: a magnifying glass, a left-pointing arrow, and a right-pointing arrow.

Result

The **Fruit and Colour program was successfully implemented in Prolog**. Multiple solutions were obtained using **backtracking** by pressing the ; operator.

Conclusion

Thus, the experiment demonstrates the use of **Prolog facts and backtracking** to retrieve multiple matching solutions. Prolog automatically searches for alternative solutions when the user requests further answers using the semicolon (;) operator.